

## OpenReview export (answer1.tex)

Rebuttal (Reviewer 1, concise for OpenReview; mathematical displays preserved)

Thanks for the constructive feedback. Q1–Q6; then Q7–Q21 (Reviewer VCqo continued, incl. Reviewer A6kZ overlaps).

**Q1.** Redraw Fig. 2 (two colors, matched FR/LR, fixed hyperparameters); momentum sweeps to appendix. Curves diagnose rank-constrained optimization; plain SGD can match momentum (MF gradient-flow view).

**Q2.** One citation style (Author–Year or numeric) throughout.

**Q3.** RF richness: frozen first layer, span of

$$\{\phi_1(\langle L^0(c_1), \cdot \rangle)\}$$

dense (e.g.  $L^2(\mu_X)$ ), replacing informal “ $\text{supp}(W^0(C_1))$ ”. Only  $\ell = 0$  needs dense-span richness;  $\ell > 1$  trains under bounded mixing

$$|W_{c,k}^{(\ell)}| \leq K, \quad \|W^{(\ell)}\|_{\infty,1} \leq rK,$$

(not full  $\mathbb{R}^d$  support for trained weights). Later mixing bounds the channel Lipschitz/coupling chain jointly with first-layer richness.

**Q4.** “Mean-field width limit” = infinite-width *mean-field* (neuronal-embedding) limit: neuron averages  $\rightarrow$  expectations, deterministic MF ODE; not NTK/lazy or  $\mu P$ .

**Q5.**

$$\phi(W^\top(Ax + c) + b) = \phi(Rx + d), \quad R := W^\top A, \quad d := W^\top c + b.$$

We keep channels (MF state  $f_k$ ) with

$$H_2(t, c_2; X, W(t)) = \sum_{k=1}^r W_{c_2,k} f_k(t; X, W(t)), \quad f_k(t; X, W) = \mathbb{E}_{C_1} [w_1(t, C_1, k) \phi_1(\langle W^0(C_1), X \rangle)].$$

Matrix form added as complement.

**Q6.** Nguyen & Pham (2023): measurable maps  $c \mapsto w(\cdot, c)$  on  $(\Omega_i, \rho_i)$ ,  $\mathbb{E}_{C_i}$  integrates  $\rho_i$ ; finite width: i.i.d. labels, empirical averages  $\rightarrow \mathbb{E}_{C_i}$  under MF scaling (pushforward / law-of-parameters). We fix vague “indices become continuous” wording.

Q7–Q21 (Reviewer VCqo).

**Q7.**

$$\mathcal{L}(W) = \mathbb{E}_Z[\mathcal{L}(Y, \hat{y}(X; W))], \quad d\mathcal{L}(Z; W) = \partial_{\hat{y}} \mathcal{L}(Y, \hat{y}(X; W)).$$

MF ODE is gradient flow of  $\mathcal{L}$  with  $\xi_\ell(t)$ . Compact defs move into Sec. 3.

**Q8.**  $\xi_\ell(t) \geq 0$  LR prefactor (constant/piecewise/exponential/cosine). Euler:

$$W_{n+1} = W_n - \varepsilon \xi_\ell(\varepsilon n) \text{ (drift at } \varepsilon n \text{)}.$$

MF limit  $\varepsilon \rightarrow 0$ ,  $t = \varepsilon n$  + width scaling  $\Rightarrow$  classical  $1/n$  LR at width  $n$ .

**Q9.** Non-degeneracy vs. trivial predictor  $\hat{y} \equiv \phi_L(0)$ :  $\mathcal{L}(W(0)) < \mathbb{E}[\mathcal{L}(Y, \phi_L(0))]$  (appendix).

**Q10.** Rank  $r$ , channels,

$$H_2(c_2; x, W) = \sum_{k=1}^r W_{c_2,k} f_k(x; W).$$

Global proof uses  $\inf_u |\phi'_2(u)| > 0$  (excludes ReLU),

$$\mathbb{E}[d\mathcal{L} \cdot \phi'_2(H_2) \mid X = x] = 0 \Rightarrow \mathbb{E}[d\mathcal{L} \mid X = x] = 0$$

(conditional gradient  $\rightarrow$  residual). ReLU: dead gates with  $\phi'_2(H_2) = 0$ ,  $d\mathcal{L} \neq 0$ . Sign-/NMF-style init may lift  $H_2$  on targets (heuristic). All- $r$  dead rare ( $e^{-O(r)}$ ); GeLU ok. Define mixing,  $\|W\|_{\infty,1}$ .

**Q11.**  $W^{(\ell)}$ :  $\|W^{(\ell)}\|_{\infty,1} \leq rK$ ; bounded-support (or same a.s. bound) laws OK.

**Q12.** Lipschitz Sec. 3.5–3.6: separate trainable  $A$  from frozen mixing norms; unify  $W/L$ .

**Q13.**  $F$ : Picard map; fixed point = trajectory.

**Q14.** Punctuation/readability.

**Q15.** Thm. 3.2: arbitrary depth (appendix); theorem will fix which widths  $\rightarrow \infty$ ; distinguish depth-2 notation in the quantitative bound.

**Q16.** Remove/rework “proof idea” to track appendix (dense span, stationarity, conditional expectation, optimality).

**Q17.**

$$B_k^{(\ell)}(t; X, W) = \mathbb{E}_{C_\ell} \left[ W_{C_\ell, k}^{(\ell)} \phi'_\ell(H_\ell(t, C_\ell; X, W)) w_\ell(t, C_\ell) \right].$$

**Q18.**  $\partial_2 \mathcal{L}$  = derivative in the prediction argument.

**Q19.** NP degeneracy (identical shift):

$$w_i^\infty(t, C_{i-1}, C_i) - w_i^\infty(0, C_{i-1}, C_i) = w_i^\#(t).$$

RF-LR breaks via

$$H_2(c_2; x, W) = \sum_{k=1}^r W_{c_2, k} f_k(x; W)$$

(distinct  $\sum_k W_{c_2, k} f_k$  per neuron); relates to diversity (Thm. 38) vs. collapse (Thm. 54, Cor. 30).

**Q20.**  $n_1, n_2$  hidden widths (depth-2 quantitative theorem); `init` indexes init laws (NP Def. 1 style). Plain “ $n_1, n_2$ , i.i.d.” wording; proof for depth 2, extends deeper.

**Q21.** “Coupling procedure” is standard finite-width embedding / NP construction (i.i.d.  $C_i(j_i)$ , empirical vs. MF means), not a separate algorithm.  $D \rightarrow D_T$  sup-norm discrepancy on  $[0, T]$ ;  $\mathcal{O}(1/\sqrt{n_{\min}} + \sqrt{\varepsilon})$  up to Grönwall; Wasserstein prose = transport view of coupled  $L^1$  integrals, not a new metric.